Database Management Systems Lab - 03

ER to Relational Schema Mapping and DDL Query Execution

Smart Hostel Management System - A Case Study

SRN: PES1UG24AM387

Contents: Task 1 - Relational schema (mapping rules, relations, schema diagram). Task 2 - DDL execution screenshots for steps 2.1 to 2.9 (SQL file: PES1UG24AM387_DDL_Lab3.sql).

Task 1 : ER to Relational Schema Mapping

1.1 Mapping rules applied

Strong entities	HOSTEL, STAFF, STUDENT, PAYMENT, RECEIPT, FACILITY each become a relation with their simple attributes; the key attribute becomes the primary key.
Weak entity ROOM (identifying relationship HAS)	ROOM includes the owner's key Hostel_ID as a foreign key; PK = (Hostel_ID, Room_Number) = owner key + partial key.
Composite attribute Student_Name	Only the simple components First_Name, Middle_Name, Last_Name are stored in STUDENT.
Derived attribute Age	Not stored; computed from Date_of_Birth when queried.
Multivalued attribute Phone_Number	New relation STUDENT_PHONE(Student_ID, Phone_Number), PK = both columns, FK to STUDENT.
Multivalued composite attribute Bed	New relation ROOM_BED holding ROOM's full key plus Bed_Number and Availability; PK = (Hostel_ID, Room_Number, Bed_Number).
1:N EMPLOYS, MAKES, PROVIDES	Foreign key of the '1' side placed on the 'N' side (STAFF.Hostel_ID, PAYMENT.Student_ID, FACILITY.Hostel_ID). Total participation of the N side gives NOT NULL.
1:N recursive SUPERVISES	STAFF.Supervisor_ID references STAFF(Staff_ID); nullable because a senior staff member has no supervisor (partial participation).
1:1 GENERATES	FK Payment_ID placed in RECEIPT (a receipt cannot exist without its payment), declared NOT NULL and UNIQUE so each payment has at most one receipt.
M:N ALLOCATES	New relation ALLOCATION with the keys of STUDENT and ROOM as FKs; PK = (Student_ID, Hostel_ID, Room_Number). The relationship attributes Allocation_Date, Check_in_Date, Check_out_Date are placed here.
M:N USES	New relation FACILITY_USAGE(Student_ID, Facility_ID), PK = both FKs.

1.2 Relational schema

Primary keys are underlined; foreign keys are listed under each relation.

HOSTEL (Hostel_ID, Hostel_Name, Location, Capacity)

STAFF (Staff_ID, Name, Salary, Designation, Hostel_ID, Supervisor_ID)

```
FK Hostel_ID → HOSTEL(Hostel_ID)
FK Supervisor_ID → STAFF(Staff_ID)
```

ROOM (Hostel_ID, Room_Number, Floor)

```
FK Hostel_ID → HOSTEL(Hostel_ID)
```

ROOM_BED (Hostel_ID, Room_Number, Bed_Number, Availability)

```
FK (Hostel_ID, Room_Number) → ROOM(Hostel_ID, Room_Number)
```

STUDENT (Student_ID, First_Name, Middle_Name, Last_Name, Gender, Date_of_Birth, Email_Address)

STUDENT_PHONE (Student_ID, Phone_Number)

```
FK Student_ID → STUDENT(Student_ID)
```

ALLOCATION (Student_ID, Hostel_ID, Room_Number, Allocation_Date, Check_in_Date, Check_out_Date)

FK Student_ID → STUDENT(Student_ID)

FK (Hostel_ID, Room_Number) → ROOM(Hostel_ID, Room_Number)

PAYMENT (Payment_ID, Payment_Date, Amount, Payment_Type, Student_ID)

FK Student_ID → STUDENT(Student_ID)

RECEIPT (Receipt_ID, Receipt_Date, Payment_ID)

FK Payment_ID → PAYMENT(Payment_ID) [UNIQUE, NOT NULL]

FACILITY (Facility_ID, Facility_Name, Operating_Hours, Hostel_ID)

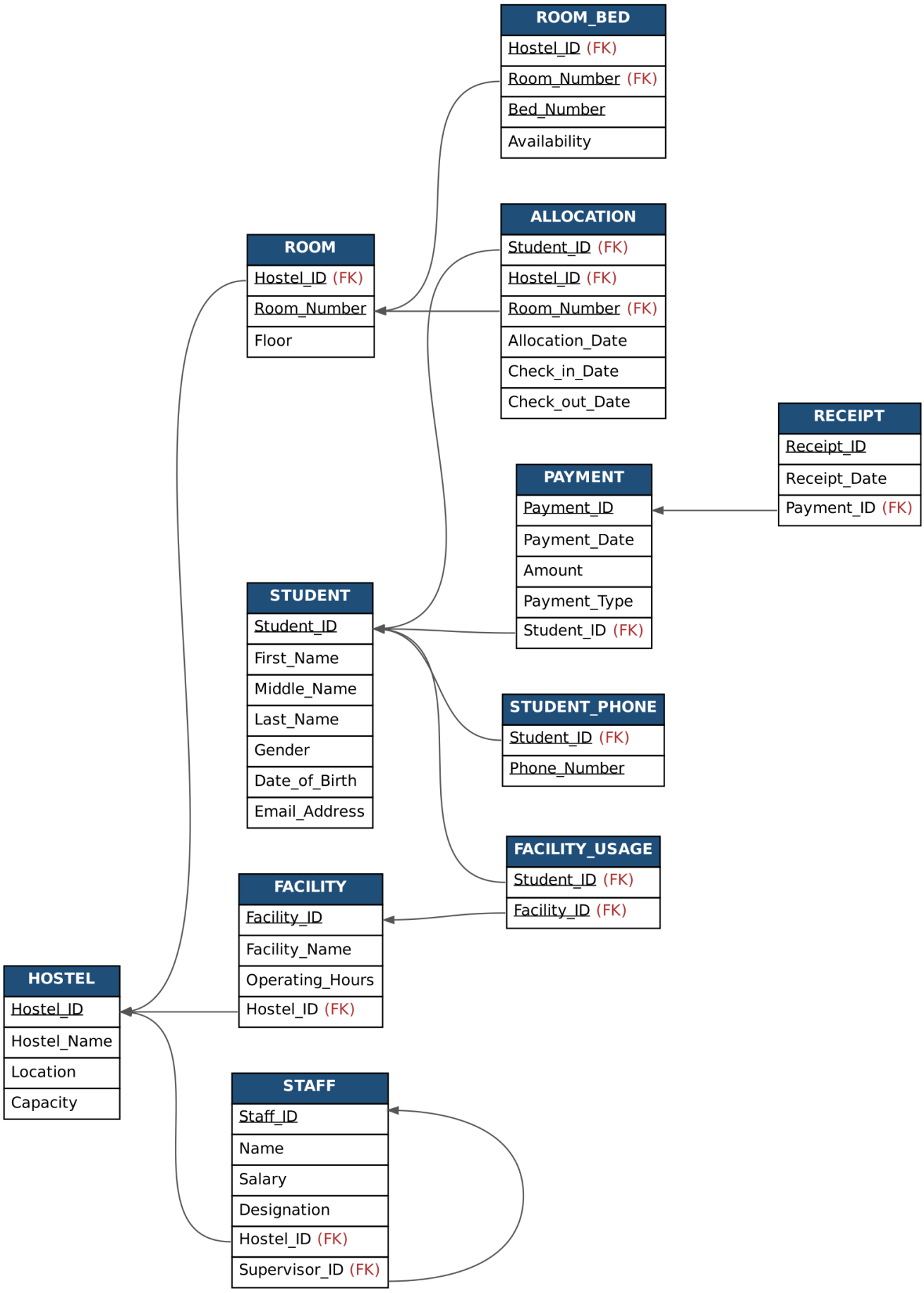
FK Hostel_ID → HOSTEL(Hostel_ID)

FACILITY_USAGE (Student_ID, Facility_ID)

FK Student_ID → STUDENT(Student_ID)

FK Facility_ID → FACILITY(Facility_ID)

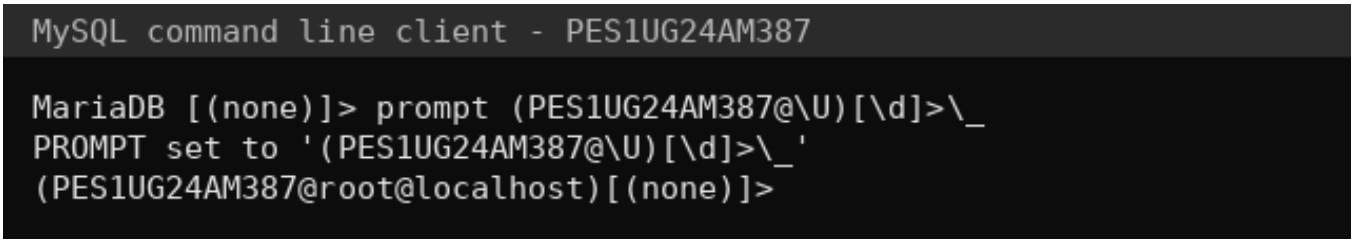
1.3 Relational schema diagram



Task 2 : DDL Query Execution (screenshots)

Setup

Prompt set to (SRN@\U)[\d]> as required before executing any command.



Step 2.1

Database smart_hostel and all 11 tables created with named PK, UNIQUE, CHECK, NOT NULL, DEFAULT and FK constraints (each FK with one referential action), followed by SHOW TABLES, DESC of every table and the list of named constraints. Note: MySQL/MariaDB always store the primary key under the name PRIMARY, so the warning 'Name pk_... ignored for PRIMARY key' is expected.

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[(none)]> DROP DATABASE IF EXISTS smart_hostel;
Query OK, 0 rows affected, 1 warning (0.000 sec)

(PES1UG24AM387@root@localhost)[(none)]> CREATE DATABASE smart_hostel;
Query OK, 1 row affected (0.000 sec)

(PES1UG24AM387@root@localhost)[(none)]> USE smart_hostel;
Database changed
(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE hostel (
    ->     Hostel_ID    VARCHAR(10)  NOT NULL,
    ->     Hostel_Name  VARCHAR(50)  NOT NULL,
    ->     Location     VARCHAR(100) NOT NULL,
    ->     Capacity     INT          NOT NULL DEFAULT 100,
    ->     CONSTRAINT pk_hostel PRIMARY KEY (Hostel_ID),
    ->     CONSTRAINT uq_hostel_name UNIQUE (Hostel_Name),
    ->     CONSTRAINT chk_hostel_capacity CHECK (Capacity > 0)
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE staff (
    ->     Staff_ID       VARCHAR(10)   NOT NULL,
    ->     Name           VARCHAR(50)   NOT NULL,
    ->     Salary         DECIMAL(10,2) NOT NULL,
    ->     Designation    VARCHAR(30)   NOT NULL DEFAULT 'Cleaner',
    ->     Hostel_ID      VARCHAR(10)   NOT NULL,
    ->     Supervisor_ID  VARCHAR(10)   NULL,
    ->     CONSTRAINT pk_staff PRIMARY KEY (Staff_ID),
    ->     CONSTRAINT chk_staff_salary CHECK (Salary > 0),
    ->     CONSTRAINT fk_staff_hostel FOREIGN KEY (Hostel_ID)
    ->         REFERENCES hostel (Hostel_ID) ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE room (
    ->     Hostel_ID    VARCHAR(10) NOT NULL,
    ->     Room_Number  VARCHAR(10) NOT NULL,
    ->     Floor        INT         NOT NULL DEFAULT 0,
    ->     CONSTRAINT pk_room PRIMARY KEY (Hostel_ID, Room_Number),
    ->     CONSTRAINT chk_room_floor CHECK (Floor >= 0),
    ->     CONSTRAINT fk_room_hostel FOREIGN KEY (Hostel_ID)
    ->         REFERENCES hostel (Hostel_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE room_bed (
    ->     Hostel_ID     VARCHAR(10) NOT NULL,
    ->     Room_Number   VARCHAR(10) NOT NULL,
    ->     Bed_Number    INT         NOT NULL,
    ->     Availability  VARCHAR(12) NOT NULL DEFAULT 'Available',
    ->     CONSTRAINT pk_room_bed PRIMARY KEY (Hostel_ID, Room_Number, Bed_Number),
    ->     CONSTRAINT chk_bed_availability CHECK (Availability IN ('Available', 'Occupied')),
    ->     CONSTRAINT fk_bed_room FOREIGN KEY (Hostel_ID, Room_Number)
    ->         REFERENCES room (Hostel_ID, Room_Number) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.005 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE student (
    ->     Student_ID     VARCHAR(15) NOT NULL,
    ->     First_Name     VARCHAR(30) NOT NULL,
    ->     Middle_Name    VARCHAR(30) NULL,
    ->     Last_Name      VARCHAR(30) NOT NULL,
    ->     Gender         CHAR(1)     NOT NULL,
    ->     Date_of_Birth  DATE        NOT NULL,
    ->     Email_Address  VARCHAR(50) NOT NULL,
    ->     CONSTRAINT pk_student PRIMARY KEY (Student_ID),
    ->     CONSTRAINT uq_student_email UNIQUE (Email_Address),
    ->     CONSTRAINT chk_student_gender CHECK (Gender IN ('M', 'F', 'O'))
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE student_phone (
    ->     Student_ID    VARCHAR(15) NOT NULL,
    ->     Phone_Number  VARCHAR(15) NOT NULL,
    ->     CONSTRAINT pk_student_phone PRIMARY KEY (Student_ID, Phone_Number),
    ->     CONSTRAINT fk_phone_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE allocation (
    ->     Student_ID       VARCHAR(15) NOT NULL,
    ->     Hostel_ID        VARCHAR(10) NOT NULL,
    ->     Room_Number      VARCHAR(10) NOT NULL,
    ->     Allocation_Date  DATE        NOT NULL,
    ->     Check_in_Date    DATE        NULL,
    ->     Check_out_Date   DATE        NULL,
    ->     CONSTRAINT pk_allocation PRIMARY KEY (Student_ID, Hostel_ID, Room_Number),
    ->     CONSTRAINT chk_alloc_checkin CHECK (Check_in_Date IS NULL OR Check_in_Date >= Allocation_Date),
    ->     CONSTRAINT chk_alloc_checkout CHECK (Check_out_Date IS NULL OR Check_out_Date >= Check_in_Date),
    ->     CONSTRAINT fk_alloc_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE,
    ->     CONSTRAINT fk_alloc_room FOREIGN KEY (Hostel_ID, Room_Number)
    ->         REFERENCES room (Hostel_ID, Room_Number) ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.007 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE payment (
    ->     Payment_ID    VARCHAR(15)   NOT NULL,
    ->     Payment_Date  DATE          NOT NULL,
    ->     Amount        DECIMAL(10,2) NOT NULL,
    ->     Payment_Type  VARCHAR(20)   NOT NULL DEFAULT 'UPI',
    ->     Student_ID    VARCHAR(15)   NOT NULL,
    ->     CONSTRAINT pk_payment PRIMARY KEY (Payment_ID),
    ->     CONSTRAINT chk_payment_amount CHECK (Amount > 0),
    ->     CONSTRAINT chk_payment_type CHECK (Payment_Type IN ('Cash', 'UPI', 'Card', 'NetBanking', 'Cheque')),
    ->     CONSTRAINT fk_payment_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.005 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE receipt (
    ->     Receipt_ID    VARCHAR(15) NOT NULL,
    ->     Receipt_Date  DATE        NOT NULL,
    ->     Payment_ID    VARCHAR(15) NOT NULL,
    ->     CONSTRAINT pk_receipt PRIMARY KEY (Receipt_ID),
    ->     CONSTRAINT uq_receipt_payment UNIQUE (Payment_ID),
    ->     CONSTRAINT fk_receipt_payment FOREIGN KEY (Payment_ID)
    ->         REFERENCES payment (Payment_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE facility (
    ->     Facility_ID      VARCHAR(10) NOT NULL,
    ->     Facility_Name    VARCHAR(40) NOT NULL,
    ->     Operating_Hours  VARCHAR(30) NOT NULL DEFAULT '06:00-22:00',
    ->     Hostel_ID        VARCHAR(10) NOT NULL,
    ->     CONSTRAINT pk_facility PRIMARY KEY (Facility_ID),
    ->     CONSTRAINT uq_facility_hostel_name UNIQUE (Hostel_ID, Facility_Name),
    ->     CONSTRAINT fk_facility_hostel FOREIGN KEY (Hostel_ID)
    ->         REFERENCES hostel (Hostel_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.005 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> CREATE TABLE facility_usage (
    ->     Student_ID   VARCHAR(15) NOT NULL,
    ->     Facility_ID  VARCHAR(10) NOT NULL,
    ->     CONSTRAINT pk_facility_usage PRIMARY KEY (Student_ID, Facility_ID),
    ->     CONSTRAINT fk_usage_student FOREIGN KEY (Student_ID)
    ->         REFERENCES student (Student_ID) ON DELETE CASCADE,
    ->     CONSTRAINT fk_usage_facility FOREIGN KEY (Facility_ID)
    ->         REFERENCES facility (Facility_ID) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected, 1 warning (0.006 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> SHOW TABLES;
+------------------------+
| Tables_in_smart_hostel |
+------------------------+
| allocation             |
| facility               |
| facility_usage         |
| hostel                 |
| payment                |
| receipt                |
| room                   |
| room_bed               |
| staff                  |
| student                |
| student_phone          |
+------------------------+
11 rows in set (0.001 sec)
```

MySQL command line client - PES1UG24AM387

```
(PES1UG24AM387@root@localhost)[smart_hostel]> DESC hostel;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Hostel_ID   | varchar(10)  | NO   | PRI | NULL    |       |
| Hostel_Name | varchar(50)  | NO   | UNI | NULL    |       |
| Location    | varchar(100) | NO   |     | NULL    |       |
| Capacity    | int(11)      | NO   |     | 100     |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC staff;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| Staff_ID      | varchar(10)   | NO   | PRI | NULL    |       |
| Name          | varchar(50)   | NO   |     | NULL    |       |
| Salary        | decimal(10,2) | NO   |     | NULL    |       |
| Designation   | varchar(30)   | NO   |     | Cleaner |       |
| Hostel_ID     | varchar(10)   | NO   | MUL | NULL    |       |
| Supervisor_ID | varchar(10)   | YES  |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
6 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC room;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Hostel_ID   | varchar(10) | NO   | PRI | NULL    |       |
| Room_Number | varchar(10) | NO   | PRI | NULL    |       |
| Floor       | int(11)     | NO   |     | 0       |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC room_bed;
+--------------+-------------+------+-----+-----------+-------+
| Field        | Type        | Null | Key | Default   | Extra |
+--------------+-------------+------+-----+-----------+-------+
| Hostel_ID    | varchar(10) | NO   | PRI | NULL      |       |
| Room_Number  | varchar(10) | NO   | PRI | NULL      |       |
| Bed_Number   | int(11)     | NO   | PRI | NULL      |       |
| Availability | varchar(12) | NO   |     | Available |       |
+--------------+-------------+------+-----+-----------+-------+
4 rows in set (0.001 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC student;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| Student_ID    | varchar(15) | NO   | PRI | NULL    |       |
| First_Name    | varchar(30) | NO   |     | NULL    |       |
| Middle_Name   | varchar(30) | YES  |     | NULL    |       |
| Last_Name     | varchar(30) | NO   |     | NULL    |       |
| Gender        | char(1)     | NO   |     | NULL    |       |
| Date_of_Birth | date        | NO   |     | NULL    |       |
| Email_Address | varchar(50) | NO   | UNI | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
7 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC student_phone;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| Student_ID   | varchar(15) | NO   | PRI | NULL    |       |
| Phone_Number | varchar(15) | NO   | PRI | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
2 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC allocation;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| Student_ID      | varchar(15) | NO   | PRI | NULL    |       |
| Hostel_ID       | varchar(10) | NO   | PRI | NULL    |       |
| Room_Number     | varchar(10) | NO   | PRI | NULL    |       |
| Allocation_Date | date        | NO   |     | NULL    |       |
| Check_in_Date   | date        | YES  |     | NULL    |       |
| Check_out_Date  | date        | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
6 rows in set (0.001 sec)
```

MySQL command line client - PES1UG24AM387

```
(PES1UG24AM387@root@localhost)[smart_hostel]> DESC payment;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| Payment_ID   | varchar(15)   | NO   | PRI | NULL    |       |
| Payment_Date | date          | NO   |     | NULL    |       |
| Amount       | decimal(10,2) | NO   |     | NULL    |       |
| Payment_Type | varchar(20)   | NO   |     | UPI     |       |
| Student_ID   | varchar(15)   | NO   | MUL | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC receipt;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| Receipt_ID   | varchar(15) | NO   | PRI | NULL    |       |
| Receipt_Date | date        | NO   |     | NULL    |       |
| Payment_ID   | varchar(15) | NO   | UNI | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
3 rows in set (0.002 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC facility;
+-----------------+-------------+------+-----+-------------+-------+
| Field           | Type        | Null | Key | Default     | Extra |
+-----------------+-------------+------+-----+-------------+-------+
| Facility_ID     | varchar(10) | NO   | PRI | NULL        |       |
| Facility_Name   | varchar(40) | NO   |     | NULL        |       |
| Operating_Hours | varchar(30) | NO   |     | 06:00-22:00 |       |
| Hostel_ID       | varchar(10) | NO   | MUL | NULL        |       |
+-----------------+-------------+------+-----+-------------+-------+
4 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC facility_usage;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Student_ID  | varchar(15) | NO   | PRI | NULL    |       |
| Facility_ID | varchar(10) | NO   | PRI | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
2 rows in set (0.001 sec)
```

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> SELECT TABLE_NAME, CONSTRAINT_NAME, CONSTRAINT_TYPE
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE CONSTRAINT_SCHEMA = 'smart_hostel'
    -> ORDER BY TABLE_NAME, CONSTRAINT_TYPE, CONSTRAINT_NAME;
+----------------+-------------------------+-----------------+
| TABLE_NAME     | CONSTRAINT_NAME         | CONSTRAINT_TYPE |
+----------------+-------------------------+-----------------+
| allocation     | chk_alloc_checkin       | CHECK           |
| allocation     | chk_alloc_checkout      | CHECK           |
| allocation     | fk_alloc_room           | FOREIGN KEY     |
| allocation     | fk_alloc_student        | FOREIGN KEY     |
| allocation     | PRIMARY                 | PRIMARY KEY     |
| facility       | fk_facility_hostel      | FOREIGN KEY     |
| facility       | PRIMARY                 | PRIMARY KEY     |
| facility       | uq_facility_hostel_name | UNIQUE          |
| facility_usage | fk_usage_facility       | FOREIGN KEY     |
| facility_usage | fk_usage_student        | FOREIGN KEY     |
| facility_usage | PRIMARY                 | PRIMARY KEY     |
| hostel         | chk_hostel_capacity     | CHECK           |
| hostel         | PRIMARY                 | PRIMARY KEY     |
| hostel         | uq_hostel_name          | UNIQUE          |
| payment        | chk_payment_amount      | CHECK           |
| payment        | chk_payment_type        | CHECK           |
| payment        | fk_payment_student      | FOREIGN KEY     |
| payment        | PRIMARY                 | PRIMARY KEY     |
| receipt        | fk_receipt_payment      | FOREIGN KEY     |
| receipt        | PRIMARY                 | PRIMARY KEY     |
| receipt        | uq_receipt_payment      | UNIQUE          |
| room           | chk_room_floor          | CHECK           |
| room           | fk_room_hostel          | FOREIGN KEY     |
| room           | PRIMARY                 | PRIMARY KEY     |
| room_bed       | chk_bed_availability    | CHECK           |
| room_bed       | fk_bed_room             | FOREIGN KEY     |
| room_bed       | PRIMARY                 | PRIMARY KEY     |
| staff          | chk_staff_salary        | CHECK           |
| staff          | fk_staff_hostel         | FOREIGN KEY     |
| staff          | PRIMARY                 | PRIMARY KEY     |
| student        | chk_student_gender      | CHECK           |
| student        | PRIMARY                 | PRIMARY KEY     |
| student        | uq_student_email        | UNIQUE          |
| student_phone  | fk_phone_student        | FOREIGN KEY     |
| student_phone  | PRIMARY                 | PRIMARY KEY     |
+----------------+-------------------------+-----------------+
35 rows in set (0.004 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.2

New first column Facility_Status with DEFAULT 'Active' added to FACILITY.

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE facility
    ->     ADD COLUMN Facility_Status VARCHAR(12) NOT NULL DEFAULT 'Active' FIRST;
Query OK, 0 rows affected (0.005 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC facility;
+-----------------+-------------+------+-----+-------------+-------+
| Field           | Type        | Null | Key | Default     | Extra |
+-----------------+-------------+------+-----+-------------+-------+
| Facility_Status | varchar(12) | NO   |     | Active      |       |
| Facility_ID     | varchar(10) | NO   | PRI | NULL        |       |
| Facility_Name   | varchar(40) | NO   |     | NULL        |       |
| Operating_Hours | varchar(30) | NO   |     | 06:00-22:00 |       |
| Hostel_ID       | varchar(10) | NO   | MUL | NULL        |       |
+-----------------+-------------+------+-----+-------------+-------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.3

The recursive SUPERVISES foreign key (fk_staff_supervisor, ON DELETE SET NULL) was left out of CREATE TABLE and is added with ALTER TABLE. The listing shows the referential action of every FK.

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE staff
    ->     ADD CONSTRAINT fk_staff_supervisor FOREIGN KEY (Supervisor_ID)
    ->         REFERENCES staff (Staff_ID) ON DELETE SET NULL;
Query OK, 0 rows affected (0.008 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC staff;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| Staff_ID     | varchar(10)  | NO   | PRI | NULL    |       |
| Name         | varchar(50)  | NO   |     | NULL    |       |
| Salary       | decimal(10,2) | NO   |     | NULL    |       |
| Designation  | varchar(30)  | NO   |     | Cleaner |       |
| Hostel_ID    | varchar(10)  | NO   | MUL | NULL    |       |
| Supervisor_ID | varchar(10)  | YES  | MUL | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
6 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> SELECT CONSTRAINT_NAME, TABLE_NAME, REFERENCED_TABLE_NAME, UPDATE_RULE, DELETE_RULE
    -> FROM information_schema.REFERENTIAL_CONSTRAINTS
    -> WHERE CONSTRAINT_SCHEMA = 'smart_hostel'
    -> ORDER BY TABLE_NAME, CONSTRAINT_NAME;
+--------------------+---------------+-----------------------+-------------+-------------+
| CONSTRAINT_NAME    | TABLE_NAME    | REFERENCED_TABLE_NAME | UPDATE_RULE | DELETE_RULE |
+--------------------+---------------+-----------------------+-------------+-------------+
| fk_alloc_room      | allocation    | room                  | CASCADE     | RESTRICT    |
| fk_alloc_student   | allocation    | student               | RESTRICT    | CASCADE     |
| fk_facility_hostel | facility      | hostel                | RESTRICT    | CASCADE     |
| fk_usage_facility  | facility_usage | facility             | RESTRICT    | CASCADE     |
| fk_usage_student   | facility_usage | student              | RESTRICT    | CASCADE     |
| fk_payment_student | payment       | student               | RESTRICT    | CASCADE     |
| fk_receipt_payment | receipt       | payment               | RESTRICT    | CASCADE     |
| fk_room_hostel     | room          | hostel                | RESTRICT    | CASCADE     |
| fk_bed_room        | room_bed      | room                  | RESTRICT    | CASCADE     |
| fk_staff_hostel    | staff         | hostel                | CASCADE     | RESTRICT    |
| fk_staff_supervisor | staff        | staff                 | RESTRICT    | SET NULL    |
| fk_phone_student   | student_phone | student               | RESTRICT    | CASCADE     |
+--------------------+---------------+-----------------------+-------------+-------------+
12 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.4

Designation renamed to Position in STAFF.

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE staff RENAME COLUMN Designation TO Position;
Query OK, 0 rows affected (0.008 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC staff;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| Staff_ID      | varchar(10)   | NO   | PRI | NULL    |       |
| Name          | varchar(50)   | NO   |     | NULL    |       |
| Salary        | decimal(10,2) | NO   |     | NULL    |       |
| Position      | varchar(30)   | NO   |     | Cleaner |       |
| Hostel_ID     | varchar(10)   | NO   | MUL | NULL    |       |
| Supervisor_ID | varchar(10)   | YES  | MUL | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
6 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.5

Gender changed from CHAR(1) to ENUM('M','F','O').

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE student MODIFY COLUMN Gender ENUM('M', 'F', 'O') NOT NULL;
Query OK, 0 rows affected (0.007 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC student;
+---------------+------------------+------+-----+---------+-------+
| Field         | Type             | Null | Key | Default | Extra |
+---------------+------------------+------+-----+---------+-------+
| Student_ID    | varchar(15)      | NO   | PRI | NULL    |       |
| First_Name    | varchar(30)      | NO   |     | NULL    |       |
| Middle_Name   | varchar(30)      | YES  |     | NULL    |       |
| Last_Name     | varchar(30)      | NO   |     | NULL    |       |
| Gender        | enum('M','F','O') | NO   |     | NULL    |       |
| Date_of_Birth | date             | NO   |     | NULL    |       |
| Email_Address | varchar(50)      | NO   | UNI | NULL    |       |
+---------------+------------------+------+-----+---------+-------+
7 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.6

Column Position dropped from STAFF.

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE staff DROP COLUMN Position;
Query OK, 0 rows affected (0.006 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC staff;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| Staff_ID      | varchar(10)   | NO   | PRI | NULL    |       |
| Name          | varchar(50)   | NO   |     | NULL    |       |
| Salary        | decimal(10,2) | NO   |     | NULL    |       |
| Hostel_ID     | varchar(10)   | NO   | MUL | NULL    |       |
| Supervisor_ID | varchar(10)   | YES  | MUL | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
5 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.7

Email_Address widened from VARCHAR(50) to VARCHAR(100).

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE student MODIFY COLUMN Email_Address VARCHAR(100) NOT NULL;
Query OK, 0 rows affected (0.012 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC student;
+--------------+------------------+------+-----+---------+-------+
| Field        | Type             | Null | Key | Default | Extra |
+--------------+------------------+------+-----+---------+-------+
| Student_ID   | varchar(15)      | NO   | PRI | NULL    |       |
| First_Name   | varchar(30)      | NO   |     | NULL    |       |
| Middle_Name  | varchar(30)      | YES  |     | NULL    |       |
| Last_Name    | varchar(30)      | NO   |     | NULL    |       |
| Gender       | enum('M','F','O') | NO   |     | NULL    |       |
| Date_of_Birth | date            | NO   |     | NULL    |       |
| Email_Address | varchar(100)    | NO   | UNI | NULL    |       |
+--------------+------------------+------+-----+---------+-------+
7 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.8

Table student renamed to students (FKs referring to it follow automatically).

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> RENAME TABLE student TO students;
Query OK, 0 rows affected (0.004 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> SHOW TABLES;
+------------------------+
| Tables_in_smart_hostel |
+------------------------+
| allocation             |
| facility               |
| facility_usage         |
| hostel                 |
| payment                |
| receipt                |
| room                   |
| room_bed               |
| staff                  |
| student_phone          |
| students               |
+------------------------+
11 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC students;
+---------------+--------------------+------+-----+---------+-------+
| Field         | Type               | Null | Key | Default | Extra |
+---------------+--------------------+------+-----+---------+-------+
| Student_ID    | varchar(15)        | NO   | PRI | NULL    |       |
| First_Name    | varchar(30)        | NO   |     | NULL    |       |
| Middle_Name   | varchar(30)        | YES  |     | NULL    |       |
| Last_Name     | varchar(30)        | NO   |     | NULL    |       |
| Gender        | enum('M','F','O') | NO   |     | NULL    |       |
| Date_of_Birth | date               | NO   |     | NULL    |       |
| Email_Address | varchar(100)       | NO   | UNI | NULL    |       |
+---------------+--------------------+------+-----+---------+-------+
7 rows in set (0.002 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```

Step 2.9

FACILITY_USAGE dropped after first dropping its constraints (fk_usage_student, fk_usage_facility and its primary key).

```
MySQL command line client - PES1UG24AM387

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE facility_usage DROP FOREIGN KEY fk_usage_student;
Query OK, 0 rows affected (0.005 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE facility_usage DROP FOREIGN KEY fk_usage_facility;
Query OK, 0 rows affected (0.006 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> ALTER TABLE facility_usage DROP PRIMARY KEY;
Query OK, 0 rows affected (0.009 sec)
Records: 0  Duplicates: 0  Warnings: 0

(PES1UG24AM387@root@localhost)[smart_hostel]> DESC facility_usage;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Student_ID  | varchar(15) | NO   |     | NULL    |       |
| Facility_ID | varchar(10) | NO   | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
2 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> DROP TABLE facility_usage;
Query OK, 0 rows affected (0.007 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]> SHOW TABLES;
+------------------------+
| Tables_in_smart_hostel |
+------------------------+
| allocation             |
| facility               |
| hostel                 |
| payment                |
| receipt                |
| room                   |
| room_bed               |
| staff                  |
| student_phone          |
| students               |
+------------------------+
10 rows in set (0.001 sec)

(PES1UG24AM387@root@localhost)[smart_hostel]>
```